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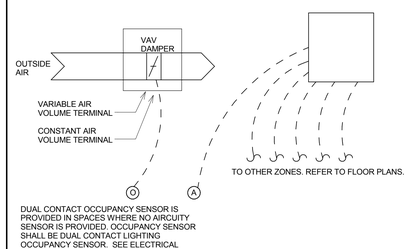
DEMAND CONTROL VENTILATION WITH CENTRALIZED CO2 SENSOR									
PROJECT DESCRIPTION	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHIC
OCCUPIED SCHEDULE	OCC_SCHED						X		YES
UNOCCUPIED SCHEDULE	UNOCC_SCHED						X		YES
VAV SET POINT MINIMUM OCCUPIED	VAV_SP_MIN_OCC					X			YES
VAV SET POINT MAXIMUM OCCUPIED	VAV_SP_MAX_OCC					X			YES
VAV SET POINT STANDBY OCCUPIED	VAV_SP_STBY_OCC					X			YES
VAV SET POINT UNOCCUPIED	VAV_SP_UNOCC					X			YES
VAV OCC SENSOR OCCUPIED	VAV_OCCS_OCC	X							
VAV OCC SENSOR UNOCCUPIED	VAV_OCCS_UNOCC	X							
VAV CFM ACTUAL	VAV_CFM_ACTUAL				X				YES
VAV DAMPER %OPEN	VAV_DAM_%				X				YES
ZONE CO2 LEVEL (PPM)	CO2_PPM				X				YES
ZONE CO2 SET POINT	CO2_SP					X			YES
CHANGE CO2 SENSOR ALARM	CO2_S_ALARM	X						BOOLEAN COMMAND FAIL	YES

ALL SPACES SERVED BY VARIABLE AIR VOLUME TERMINAL OR CONSTANT AIR VOLUME TERMINAL



DUAL CONTACT OCCUPANCY SENSOR IS PROVIDED IN SPACES WHERE NO AIRCUITY SENSOR IS PROVIDED. OCCUPANCY SENSOR SHALL BE DUAL CONTACT LIGHTING OCCUPANCY SENSOR. SEE ELECTRICAL SPECS.

DEMAND CONTROL VENTILATION WITH CENTRALIZEDCO2 SENSOR(AIRCUITY)

1. DEMAND CONTROL VENTILATION SEQUENCE OF OPERATION (BASIS OF DESIGN: AIRCUITY)

1.0 OCCUPIED MODE:

1.1. Aircuity FMS shall monitor CO2 levels in each zone.

1.2. Aicuity FMS shall compare the CO2 levels in each zone with OA CO2. A differential CO2 signal shall be sent to the BMS. The signal shall be a 0-10 VDC signal where 0 VDC represents 200 PPM differential, 5 VDC represents 500 PPM and 10 VDC represents 800 PPM differential.

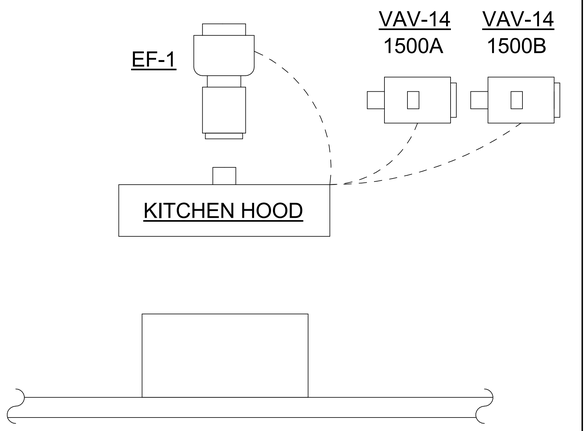
1.3. The VAV box controller shall modulate the VAV box damper to maintain as follows: 200 PPM (adj.) damper is closed, 500 PPM damper is half open (adj.), and 800 PPM (adj.) damper is full open.

1.4. Controls contractor is responsible for providing field mounted VAV box controller to accomplish sequence and achieve all listed control points.

1.5 If a pace is not provided with an aircuity sensor, VAV box shall respond to occuapancy sensor. Durring occpied hours the VAV box shall have an occpied standby CFM setting that will enable when NO occupancy is detected and an occupied CFM setting that will enable when occupancy is detected. If VAV box serves more than one space then all space occupany sensors must detect NO occupancy before VAV box is allowed to enable occupied standyby CFM setting.

MECHANICAL CONTROL LEGEND

AFF	ABOVE FINISHED FLOOR	Ts	
BAS	BUILDING AUTOMATION SYSTEM	H	
CO_2	CARBON DIOXIDE	LL	
TCC	TEMPERATURE CONTROL CONTRACTOR	P	
DP	DEWPOINT	EP	
EA	EXHAUST AIR PATH	DPSW	
RA	RETURN AIR PATH	DPS	
SA	SUPPLY AIR PATH	C	
HPS/R	HEAT PUMP WATER SUPPLY/RETURN	M	
NC	NORMALLY CLOSED	F	
OA	OUTSIDE AIR PATH	CS	
OCC	OCCUPANCY	SD	AVERAGING TEMPERATURE SENSOR
PRESS	PRESSURE	COS	AVERAGING TEMPERATURE SENSOR
DI	DIGITAL INPUT	DSP-HL	AVERAGING TEMPERATURE SENSOR
DO	DIGITAL OUTPUT	DSP-LL	AVERAGING TEMPERATURE SENSOR
AI	ANALOG INPUT	ZN-DP	AVERAGING TEMPERATURE SENSOR
AO	ANALOG OUTPUT	ZN-CO2	AVERAGING TEMPERATURE SENSOR
VFD	VARIABLE FREQUENCY DRIVE	ZN-OCC	AVERAGING TEMPERATURE SENSOR
RH	RELATIVE HUMIDITY	ZN-T	AVERAGING TEMPERATURE SENSOR
MAU	MAKE-UP AIR UNIT	CO2	AVERAGING TEMPERATURE SENSOR
O/H	OCCUPIED HEATING SETPOINT		CENTRIFUGAL FAN
O/C	OCCUPIED COOLING SETPOINT		
U/H	UNOCCUPIED HEATING SETPOINT		
U/C	UNOCCUPIED COOLING SETPOINT		
K	EMERGENCY HVAC/VENTILATION KILL BUTTON		
Ta	AVERAGING TEMPERATURE SENSOR		



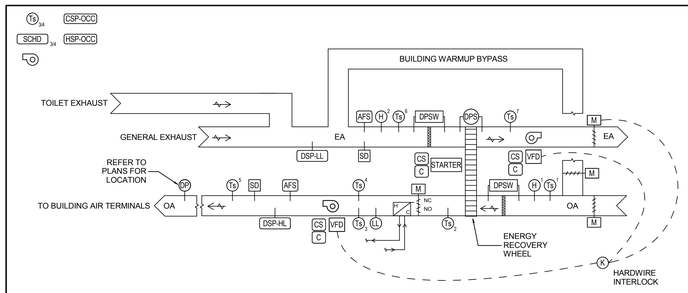
KITCHEN HOOD & EXHAUST CONTROL SEQUENCE

NOT TO SCALE

KITCHEN HOOD & EXHAUST CONTROL SEQUENCES:

1. The kitchen exhuast hood shall be interlocked with Exhuast Fan EF-1, Gym Air Terminal Box VAV-14 1500A and VAV-14 1500B, and normally open Gymnasium 1500 motorized exhaust damper serving louver L-2.
2. Upon activation of EF-1, VAV-14 1500A and 1500B shall both be activated and each set to a minimum of 1500 CFM. Both boxes shall return to normal operation upon the deactivation of exhaust fan EF-1.
3. Upon activation of EF-1 Normally open Gymnasium 1500 motorized exhaust damper shall be energized closed so that gym over pressurizes and transfers air for make-up into the kitchen. Exhaust damper shall return to normal position upon deactivation of EF-1.

KITCHEN HOOD & EXHAUST CONTROL POINTS									
POINT DESCRIPTION	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHICS
KITCHEN HOOD STATUS	DW_S	X							X
EXHAUST FAN COMMAND	DW_EF_C		X						X
VAV BOX COMMAND	DW_VAV_C		X						X
MOTORIZED EXHAUST DAMPER	EA_D_C		X						X
MOTORIZED EAXHAUST DAMPER STATUS	EA_D_S	X							X



OUTSIDE AIR SYSTEM (OA-1) SCHEMATIC

NOT TO SCALE

OUTSIDE AIR SYSTEMS:

1. THERE ARE TWO OUTSIDE AIR UNITS IN THIS FACILITY.
2. EACH SYSTEM SHALL OPERATE UNDER THE CONTROL OF A LOCAL, STAND-ALONE MICROPROCESSOR BASED DDC CONTROLLER.
3. EACH SYSTEM SHALL BE PLACED INTO THE OCCUPIED/UNOCCUPIED MODE BASED UPON THE USER ADJUSTABLE SCHEDULE AT THE NETWORK CONTROLLER. THESE SYSTEMS SHALL BE IN THE OCCUPIED MODE DURING REGULAR SCHOOL HOURS ONLY. SEE BUILDING OCCUPANCY SCHEDULE.
4. IF COMMUNICATION IS LOST BETWEEN THE NETWORK CONTROLLER AND THE OUTISDE AIR SYSTEM CONTROLLER. THEN THE OUISIDE AIR SYSTEM SHALL BE PLACED INTO THE UNOCCUPIED MODE UNTIL COMMUNICATION IS RESTORED.
5. EACH SYSTEM WILL BE PLACED INTO A MODE OF OPERATION BASED UPON THE FOLLOWING ADJUSTABLE TERMPERATURE SCHEDULE:

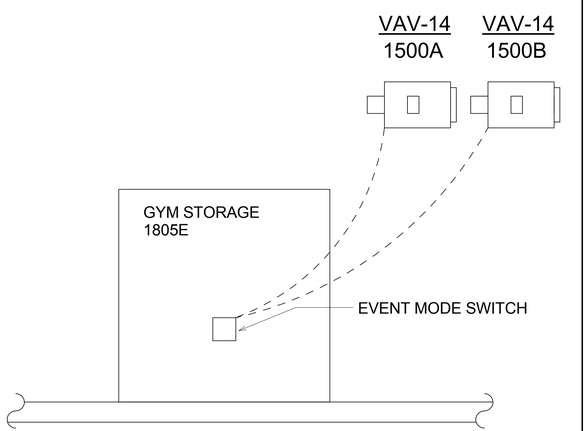
OUTSIDE AIR TEMPERATURE	MODE OF OPERATION
75 DEG F OR GREATER (ADJ.)	COOLING MODE
BETWEEN 50 DEG F AND 75 DEG F (ADJ.)	ECONOMIZER MODE
50 DEG F OR LESS (ADJ.)	HEATING MODE

6. IN THE OCCUPIED MODE OF FREEZESTAT (LOW LIMIT) MODE:
   - THE SUPPLY FAN AND EXHAUST SHAL BE OFF.
   - THE ENERGY RECOVERY WHEEL SHALL BE OFF.
   - THE OUTSIDE AIR DAMPER AND EXHAUST AIR DAMPER SHALL BE FULLY CLOSED,
   - FACE AND BYPASS DAMPER SHALL BE IN FULL FACE POSITION.
7. WHEN PLACED INTO THE OCCUPIED MODE, THE FOLLOWING SHALL OCCUR IN SEQUENTIAL ORDER AFTER OPERATION OF THE WATER-TO-WATER HEAT PUMP SYSTEM HAS BEEN PROVED:
   - THE ENERGY RECOVERY WHEEL SHALL START AND OPERATION SHALL BE PROVED VIA CURRENT SWITCH,
   - THE OUTSIDE AIR DAMPER AND EXHAUST AIR DAMPER SHALL FULLY OPEN AND BE PROVED VIA CURRENT SWITCH,
   - THE SUPPLY FAN/VFD AND EXHAUST FAN/VFD SHALL START AND OPERATION SHALL BE PROVED VIA CURRENT SWITCH,
   - THE SYSTEM SHALL NOT START IF ANY ONE COMPONENT DOES NOT PROVE OPERATION, INCLUDING THE HYDRONIC WATER-TO-WATER HEAT EXCHANGER SYSTEM.
8. IN THE OCCUPIED MODE, THE FACE AND BYPASS DAMPERS SHALL MODULATE TO MAINTAIN DISCHARGE AIR TEMPERATURE (ADJ.) BASED UPON THE FOLLOWING SCHEDULE:

DISCHARGE AIR TEMPERATURE	MODE OF OPERATION
68 DEG F (ADJ.)	COOLING MODE
VARIES	ECONOMIZER MODE
65 DEG F (ADJ.)	HEATING MODE

(NOTE THAT THERE SHOULD BE NOT BE MORE THAN APPROXIMATELY 60% AIRFLOW NECESSARY ACROSS THE COOLING/HEATING COIL TO MAINTAIN REQUIRED DISCHARGE AIR TEMPERATURE. SET DAMPER ACTUATOR CONTROL ACCORDINGLY.)

9. IF THE OUTSIDE AIR TEMPERATURE IS BETWEEN 60 DEG F (ADJ.) AND 65 DEG F (ADJ.), THEN THE ENERGY RECOVERY WHEEL SHALL BE OFF. THE WATER - TO -WATER HEAT PUMP SYSTEM SHALL REMAIN OFF.
10. DEHUMIDIFICATION MODE: WHEN THE DUCT-MOUNTED EXHAUST AIR RELATIVE HUMIDITY SENSOR READS 65% OR GREATER, ENERGY RECOVERY WHEEL SHALL REMAIN ON AND BYPASS DAMPER SHALL CLOSE ALLOWING THE UNIT TO OPERATE WITH HALF AIRFLOW UNTIL EXHAUST AIR RELATIVE HUMIDITY IS BELOW 60%.
11. THE SUPPLY FAN SHALL BE CONTROLLED BY THE VFD BASED ON FIELD-MOUNTED DUCT PRESSURE SETPOINT. FAN SUPPLY STATIC PRESSUREOPTIMIZATION SHALL BE UTILIZED BY POLLING OF ASSOCIATED VAVICAV BOX DAMPER POSITIONS AND RESETTING OF PRESSURE SETPOINT AIALLOWED. DDC SYSTEM DETERMINES VAV BOX WITH GREATEST DAMPER OPEN POSITION ONCE EVERY TEN MINUTES. UNIT'S SUPPLY AIR STATPRESSURE SETPOINT DECREASED BY 0.1" WG AND 1.25" WG. SETPOINT IS 0.75" WG WHEN UNIT IS FIRST STARTED. DETERMINE IN THE FIELD WITH THE TAB CONTRCTOR THE STATIC PRESSURE SETPOINT TO OBTAIN ACCEPTABLE AIRFLOW AS DESIGNED.
12. HARDWARE INTERLOCK SUPPLY FAN, EXHAUST FAN, AND ISOLATION DAMPERS WITH EMERGENCY HVAC/VENTILATION KILL BUTTON.
13. THE EXHAUST FAN SHALL TRACK THE OUTSIDE AIR FAN V.A AIRFLOW STATION AND SHALL MAINTAIN A 1,000 CFM OFFSET FOR EACH UNIT TO MAINTAIN A POSITIVE BUILDING PRESSURIZATION. ALL RESTROOMS ARE PROVIDED WITH A CONSTANT VOLUME BOX AND SHALL MAINTAIN THE CFM INDICATED ON THE DRAWINGS. IF WHILE THE EXHAUST FAN TRACKS THE OUTISDE AIR FAN THE CFM IN THE RESTROOM DROPS BELOW THIS AIRFLOW IN ANY ROOM BY 10%, THEN THE OUTSIDE AIR SHALL STOP DECREASING TO MAINTAIN APPROPRIATE TOILET EXHAUST AIR FLOW.
14. THE COIL IS A FULL FLOW COIL WITH FACE AND BYPASS DAMPER CONTROL FOR FREEZE PROTECTION AND DEHUMIDIFICATION CONTROL. LOW TEMPERATURE CUTOUT IS LOCATED ON THE DISCHARGE OF THE COIL AND SHALL ACTIVATE WHEN THE LEAVING AIR TEMPERATURE FALLS BELOW 35 DEG F. THE SUPPLY AND EXHAUST FANS SHALL STOP, OUTSIDE AIR AND EXHAUST AIR DAMPERS SHALL CLOSE.
15. LOW TEMPERATURE SENSOR ERW FROST ALARM LOCATED ON THE DISCHARGE OF THE ENERGY RECOVERY WHEEL, WHEN THE LEAVING AIR TEMPERATURE FALLS BELOW 40 DEG F THE SUPPLY AND EXHAUST FANS SHALL STOP, OUTSIDE AND EXHAUST AIR DAMPERS SHALL CLOSE. WHEN TEMPERATURE RAISES TO 45 DEG F (ADJ.) SYSTEM SHALL RETURN TO NORMAL OPERATION.
16. A MANUAL RESET LOW LIMIT INSTALLED DOWNSTREAM OF THE CHILLED/HOT WATER COIL SHALL STOP THE OPERATION OF THE SYSTEM IF THE DISCHARGE TERMPATURE FALLS ` BELOW 35 DEG F.
17. A SMOKE DETECTOR SHALL BE LOCATED IN EACH AIR SYSTEM. IF SMOKE IS DETECTED, THEN THE SYSTEM SHALL SHUTOFF AND AN AUDIO/VISUAL ALARM SHALL ACTIVATE. UPON CORRECTION OF PROBLEM, THE SYSTEM SHALL BE RESET AND SHALL RETURN TO NORMAL OPERATION. COORDINATE WITH FIRE ALARM SYSTEM. ANY THEIVATION OF BUILDING'S FIRE ALARM SYSTEM SHALL SHUTDOWN THE OA UNITS COMPLETELY.
18. THE UNIT SHALL BE PROVIDED WITH A BUILDING WARM UP/COOL DOWN BYPASS DUCT. THIS BYPASS SHALL BYPASS THE WHEEL AND CREATE A RECIRCULATING SYSTEM AND NOT USE ANY FRESH AIR SINCE THE BUILDING IS UNOCCUPIED. IF THE TEMPERATURE HAS FALLEN BELOW SETPONT THE FANS SHALL BE ENGAGED AND THE UNIT SHALL DISCHARGE 95 DEG F (ADJ.). THE ASSOCIATED VAV BOX IN EACH ZONE SHALL OPEN USING THE OUTSIDE AIR UNIT TO BOOST TEMPERATURE. CONVERSELY, IF THE TEMPERATURE RAISES ABOVE UNOCCUPIED SETPOINT, THE UNIT SHALL CONTROL TO A 55 DEG F (ADJ.) DISCHARGE AIR TEMPERATURE TO REDUCE THE UNOCCUPIED TEMPERATURE. THE ASSOCIATED VAV BOX IN EACH ZONE SHALL OPEN. ALL CONSTANT VOLUME TOILET EXHAUST BOXES SHALL BE CLOSED DURING WARMUP/COOL DOWN CYCLE.



NOT TO SCALE

GYMNASIUM OUTSIDE AIR CONTROL SEQUENCES:

1. GYMNASIUM 1800 **normal mode** of operation shall be such that outside air is provided for classroom use. GYMNASIUM 1800 normal minimum shall be 1200 CFM and normal maximum shall 1600 CFM. VAV-box A serving gym shall maintain CFM values and modulate according to CO2.
2. GYMNASIUM 1800 **event mode** shall be initiated by a push button in Gym Storage 1805E and start a 2 hour timer (adj.) for an event mode which will automically disable upon experation of timer. GYMNASIUM 1800 Event mode shall have a minimum CFM value of 1600 CFM and maximum value of 3000 CFM. VAV box A and B shall work together and modulate based on CO2 to maintain CFM values.
3. Note that Kitchen hood operation shall overide both of the above modes and take controls of VAV-boxes A and B serving gym.

OA-1 POINTS LIST

DISPLAY NAME	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHIC
BUILDING STATIC PRESSURE	BLDG_SP				X			OUT OF RANGE	YES
DUAL TEMP MODULATING 2-WAY %OPEN	DT_CV-1_PO					X		BOOLEAN COMMAND FAIL	YES
DUAL TEMP MODULATING 2-WAY OVERIDE	DT_CV-1_OVRD						X		
DEHUMIDIFICATION SETPOINT	OA_DHUM_SPT						X		YES
OA INTAKE HUMIDITY SENSOR 1	OA_HUM_S_1				X				
EA DISCHARGE HUMIDITY SENSOR 2	EA_HUM_S_2				X				
OA_EA BYPASS DAMPER	OA_EA_BYP_DAMP_C					X		BOOLEAN COMMAND FAIL	
OA_EA BYPASS DAMPER STATUS	OA_EA_BYP_DAMP_ST	X							
OA_EA BYPASS DAMPER OVERIDE	OA_EA_BYP_DAMP_OVRD						X		YES
OA INTAKE DAMPER	OA_INT_DAMP_C					X		BOOLEAN COMMAND FAIL	YES
OA INTAKE DAMPER STATUS	OA_INT_DAMP_ST	X							YES
OA INTAKE DAMPER OVERIDE	OA_INT_DAMP_OVRD						X		YES
EA RELIEF DAMPER	EA_INT_DAMP_C					X		BOOLEAN COMMAND FAIL	YES
EA RELIEF DAMPER STATUS	EA_INT_DAMP_ST	X							YES
EA RELIEF DAMPER OVERIDE	EA_INT_DAMP_OVRD						X		YES
OUTSIDE AIR TEMPERATURE SENSOR 1	OA_TEMP_S_1				X				YES
OUTSIDE AIR TEMPERATURE SENSOR 2	OA_TEMP_S_2				X				YES
OUTSIDE AIR H/C TEMPERATURE SENSOR 3	OA_TEMP_S_3				X				YES
OUTSIDE AIR BYPASS TEMPERATURE SENSOR 4	OA_TEMP_S_4				X				YES
OUTSIDE AIR TEMPERATURE SENSOR 5	OA_TEMP_S_5				X				
OUTSIDE AIR T3/T4 AVERAGE TEMPERATURE	OA_TEMP_S_AVG_3&4						X		
OA PREFILTER STATUS	OA_FLTR_DP_S				X			OUT OF RANGE	
OA INTAKE HUMIDITY SENSOR 1	OA_RH_S				X				YES
MODULATING FACE AND BYPASS DAMPER	OA_FBYP_DAMPER					X			YES
LOW LIMIT FREEZE STAT	LL-FRZ_STAT	X						OUT OF RANGE	YES
SCHEDULE	OCC_S			X					
OA FANS START/STOP	OAFAN_RUN_C		X					BOOLEAN COMMAND FAIL	YES
OA FANS START/STOP OVERRIDE	OAFAN_RUN_OVRD			X				BOOLEAN COMMAND FAIL	YES
OA FANS STATUS	OAFAN_RUN_ST	X						BOOLEAN COMMAND FAIL	YES
OA FANS VFD FAULT	OAFAN_VFDFLT_S	X						BOOLEAN CHANGE OF STATUS	YES
OA FANS VFD SPEED	OAFAN_VFD_SPD_C					X			YES
OA FANS VFD SPEED OVERRIDE	OAFAN_VFD_SPD_OVRD						X		YES
FAN 1 OA AIR CFM	OAFAN1_AF_CFM				X				YES
FAN 2 OA AIR CFM	OAFAN2_AF_CFM				X				YES
OA TOTAL CFM	OA_AF_CFM_TOTAL				X				YES
OA SMOKE DETECTOR	OA_SMOKE_DET	X						BOOLEAN CHANGE OF STATUS	YES
OA STATIC PRESSURE	OA_SP_S				X				YES
OA STATIC PRESSURE SETPOINT	OA_SP_SETP						X		YES
OA STATIC PRESSURE RESET SETPOINT	OA_SP_RESET_SETP						X		YES
OCCUPIED COOLING SETPOINT	OCC_CLG_SETP						X		YES
OCCUPIED HEATING SETPOINT	OCC_HTG_SETP						X		YES
DUCT STATIC PRESSURE HIGH LIMIT	OA_DSP_HL	X						OUT OF RANGE	YES
DUCT STATIC PRESSURE LOW LIMIT	EA_DSP_LL	X						OUT OF RANGE	
UNOCCUPIED BUILDING WARMUP HEATING SETPOINT	UNOCC_WARMUP_HTG_SETP						X		
EA SMOKE DETECTOR	EA_SMOKE_DET	X						BOOLEAN COMMAND FAIL	
EXHAUST AIR TEMPERATURE SENSOR 6	EA_TEMP_S_6				X				
EXHAUST AIR TEMPERATURE SENSOR 7	EA_TEMP_S_6				X				YES
EA PREFILTER STATUS	EA_FLTR_DP_S				X			OUT OF RANGE	YES
ENERGY RECOVERY WHEEL DIFF. PRESS. SENSOR	ERW_DP_S				X			OUT OF RANGE	
ENERGY RECOVERY WHEEL START/STOP	ERW_RUN_C		X					BOOLEAN COMMAND FAIL	
ENERGY RECOVERY WHEEL START/STOP OVERIDE	ERW_RUN_OVRD			X				BOOLEAN COMMAND FAIL	
ENERGY RECOVERY WHEEL STATUS	ERW_RUN_ST	X						BOOLEAN COMMAND FAIL	
ENERGY RECOVERY WHEEL FREEZE CONTROL	ERW_FRZ_C			X					
EA FANS START/STOP	EAFAN_RUN_C							BOOLEAN COMMAND FAIL	
EA FANS START/STOP OVERRIDE	EAFAN_RUN_OVRD							BOOLEAN COMMAND FAIL	
EA FANS STATUS	EAFAN_RUN_ST							BOOLEAN COMMAND FAIL	
EA FANS VFD FAULT	EAFAN_VFDFLT_S							BOOLEAN CHANGE OF STATUS	
EA FANS VFD SPEED	EAFAN_VFD_SPD_C								
EA FANS VFD SPEED OVERRIDE	EAFAN_VFD_SPD_OVRD								
FAN 1 EA AIR CFM	EAFAN1_AF_CFM								
FAN 2 EA AIR CFM	EAFAN2_AF_CFM								
EA TOTAL CFM	EA_AF_CFM_TOTAL								
SUM OF CONNECTED TERMINAL UNITS	TERM_UNIT_CFM_TOTAL				X				

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CONTROL SCHEMATICS
HOLABIRD ACADEMY PK-8
CITY OF BALTIMORE, MARYLAND

DATE	DESCRIPTION

M-8.0

03/13/2017
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